

Hossein Hadipour

Curriculum Vitae



Education

- 2025–Present **Postdoc in Computer Science**, *Ruhr University Bochum*, Bochum, Germany.
- 2021–2024 **Ph.D. in Computer Science**, *Graz University of Technology*, Graz, Austria.
 - Thesis: Automated Methods in Design and Analysis of Symmetric-Key Primitives
 - Supervisor: Dr. Maria Eichlseder
- 2014–2016 **Master of Pure Mathematics**, *University of Tehran*, Tehran, Iran.
 - Thesis: Topics in Algebraic Attacks on Cryptosystems
 - Supervisor: Dr. Hosein Sabzrou
- 2012–2016 **Bachelor of Electrical Engineering**, *K. N. Toosi University of Technology*, Tehran, Iran.
 - Project 1: Evaluation of Algebraic Attacks on Cryptosystems
 - Project 2: Implementation of RFID Access Control System via AVR Microcontrollers
 - Supervisor: Dr. Bahareh Akhbari
- 2010–2014 **Bachelor of Applied Mathematics**, *K. N. Toosi University of Technology*, Tehran, Iran.
 - Supervisor: Prof. A. Reza Moghaddamfar

Research and Publications

- Shuo Peng, Akram Khalesi, Zahra Ahmadian, **Hossein Hadipour**, Jiahui He, Kai Hu, Zhongfeng Niu, Shahram Rasoolzadeh, and Meiqin Wang. “Cube and Integral Attacks on ChiLow-32”. In: *IACR Trans. Symmetric Cryptol.* 2026.1 (2026), pp. 253–292. DOI: [10.46586/TOSC.V2026.I1.253-292](https://doi.org/10.46586/TOSC.V2026.I1.253-292)
- Cihangir Tezcan, Gregor Leander, and **Hossein Hadipour**. “Cryptanalysis: Theory Versus Practice Correcting Cryptanalysis Results on Ascon, ChaCha, and Serpent Using GPUs”. In: *IACR Trans. Symmetric Cryptol.* 2025.3 (2025), pp. 729–754. DOI: [10.46586/TOSC.V2025.I3.729-754](https://doi.org/10.46586/TOSC.V2025.I3.729-754)
- Hossein Hadipour**, Yosuke Todo, Mostafizar Rahman, Maria Eichlseder, Ravi Anand, and Takanori Isobe. “Turning Multiple Key-Dependent Attacks into Universal Attacks”. In: *IACR Cryptol. ePrint Arch.* 2025 (2025), p. 1996. URL: <https://eprint.iacr.org/2025/1996>
- Chengcheng Chang, **Hossein Hadipour**, Kai Hu, Muzhou Li, and Meiqin Wang. “Mix-Basis Geometric Approach to Boomerang Distinguishers”. In: *IACR Trans. Symmetric Cryptol.*

2025.3 (2025), pp. 693–728. DOI: [10.46586/TOSC.V2025.I3.693-728](https://doi.org/10.46586/TOSC.V2025.I3.693-728). URL: <https://doi.org/10.46586/tosc.v2025.i3.693-728>

- Debasmita Chakraborty, **Hosein Hadipour**, Anup Kumar Kundu, Mostafizar Rahman, Prathamesh Ram, Yu Sasaki, Dilip Sau, and Aman Sinha. “Breaking the Twinkle Authenticated Encryption Scheme and Analyzing Its Underlying Permutation”. In: *Selected Areas in Cryptography - SAC 2025 - 32nd International Conference, Toronto, ON, Canada, August 13-15, 2025, Revised Selected Papers*. Ed. by Christina Boura, Atefeh Mashatan, and Ali Miri. Vol. 16207. Lecture Notes in Computer Science. Springer, 2025, pp. 79–109. DOI: [10.1007/978-3-032-10536-3_4](https://doi.org/10.1007/978-3-032-10536-3_4)
- **Hosein Hadipour**, Patrick Derbez, and Maria Eichlseder. “Revisiting Differential-Linear Attacks via a Boomerang Perspective with Application to AES, Ascon, CLEFIA, SKINNY, PRESENT, KNOT, TWINE, WARP, LBlock, Simeck, and SERPENT”. in: *Advances in Cryptology - CRYPTO 2024*. Vol. 14923. LNCS. Springer, 2024, pp. 38–72. DOI: [10.1007/978-3-031-68385-5_2](https://doi.org/10.1007/978-3-031-68385-5_2)
- Debasmita Chakraborty, **Hosein Hadipour**, Phuong Hoa Nguyen, and Maria Eichlseder. “Finding Complete Impossible Differential Attacks on AndRX Ciphers and Efficient Distinguishers for ARX Designs”. In: *IACR Trans. Symmetric Cryptol.* 2024.3 (2024), pp. 84–176. DOI: [10.46586/TOSC.V2024.I3.84-176](https://doi.org/10.46586/TOSC.V2024.I3.84-176)
- **Hosein Hadipour**, Sadegh Sadeghi, and Maria Eichlseder. “Finding the Impossible: Automated Search for Full Impossible-Differential, Zero-Correlation, and Integral Attacks”. In: *Advances in Cryptology - EUROCRYPT 2023*. Ed. by Carmi Hazay and Martijn Stam. Vol. 14007. LNCS. Springer, 2023, pp. 128–157. DOI: [10.1007/978-3-031-30634-1_5](https://doi.org/10.1007/978-3-031-30634-1_5)
- **Hosein Hadipour** and Yosuke Todo. “Cryptanalysis of QARMAv2”. In: *IACR Trans. Symmetric Cryptol.* 2024.1 (2024), pp. 188–213. DOI: [10.46586/TOSC.V2024.I1.188-213](https://doi.org/10.46586/TOSC.V2024.I1.188-213)
- **Hosein Hadipour**, Simon Gerhalter, Sadegh Sadeghi, and Maria Eichlseder. “Improved Search for Integral, Impossible Differential and Zero-Correlation Attacks Application to Ascon, ForkSKINNY, SKINNY, MANTIS, PRESENT and QARMAv2”. In: *IACR Trans. Symmetric Cryptol.* 2024.1 (2024), pp. 234–325. DOI: [10.46586/TOSC.V2024.I1.234-325](https://doi.org/10.46586/TOSC.V2024.I1.234-325)
- **Hadipour, Hosein**, Marcel Nageler, and Maria Eichlseder. “Throwing Boomerangs into Feistel Structures: Application to CLEFIA, WARP, LBlock, LBlock-s and TWINE”. in: *IACR Trans. Symmetric Cryptol.* 2022.3 (2022), pp. 271–302. DOI: [10.46586/tosc.v2022.i3.271-302](https://doi.org/10.46586/tosc.v2022.i3.271-302)
- **Hosein Hadipour** and Maria Eichlseder. “Integral Cryptanalysis of WARP based on Monomial Prediction”. In: *IACR Trans. Symmetric Cryptol.* 2022.2 (2022), pp. 92–112. DOI: [10.46586/tosc.v2022.i2.92-112](https://doi.org/10.46586/tosc.v2022.i2.92-112)
- **Hosein Hadipour** and Maria Eichlseder. “Autoguess: A Tool for Finding Guess-and-Determine Attacks and Key Bridges”. In: *ACNS*. vol. 13269. LNCS. Springer, 2022, pp. 230–250. DOI: [10.1007/978-3-031-09234-3_12](https://doi.org/10.1007/978-3-031-09234-3_12)
- Hadi Soleimany, Nasour Bagheri, **Hosein Hadipour**, Prasanna Ravi, Shivam Bhasin, and Sara Mansouri. “Practical Multiple Persistent Faults Analysis”. In: *IACR Trans. Cryptogr. Hardw. Embed. Syst.* 2022.1 (2022), pp. 367–390. DOI: [10.46586/tches.v2022.i1.367-390](https://doi.org/10.46586/tches.v2022.i1.367-390)
- **Hosein Hadipour** and Nasour Bagheri. “Improved Rectangle Attacks on SKINNY and CRAFT”. in: *IACR Trans. Symmetric Cryptol.* 2021.2 (2021), pp. 140–198. DOI: [10.46586/tosc.v2021.i2.140-198](https://doi.org/10.46586/tosc.v2021.i2.140-198)
- **Hosein Hadipour**, Sadegh Sadeghi, Majid M. Niknam, and Nasour Bagheri. “Comprehensive security analysis of CRAFT”. in: *IACR Trans. Symmetric Cryptol.* 2019.4 (2019), pp. 290–317. DOI: [10.13154/tosc.v2019.i4.290-317](https://doi.org/10.13154/tosc.v2019.i4.290-317)

Reviews

- Reviewer for [FSE 2027/ToSC 2026-2~ToSC 2027-1](#)
- Reviewer for [ASIACRYPT 2026](#)
- Artifact Review Chair at [FSE 2026/ToSC](#)
- Reviewer for [CASCADE 2026](#)
- Reviewer for [Designs, Codes and Cryptography \(DCC\) 2025](#)
- Subreviewer for [ASIACRYPT 2025](#)
- Subreviewer for [CRYPTO 2025](#)
- Subreviewer for [Selected Area in Cryptography \(SAC\) 2025](#)
- Artifact review committee at [ASIACRYPT 2024](#)
- Subreviewer for [ASIACRYPT 2024](#)
- Subreviewer for [EUROCRYPT 2024](#)
- Subreviewer for [EUROCRYPT 2023](#)
- Subreviewer for [ASIACRYPT 2023](#)
- Subreviewer for [EUROCRYPT 2023](#)
- Subreviewer for [CRYPTO 2022](#)
- Subreviewer for [ASIACRYPT 2022](#)
- Reviewer for [IET Information Security 2022](#)
- Reviewer for [Designs, Codes and Cryptography \(DCC\) 2022](#)

Visits

- May 2024 [Inria](#), Paris, France ([Visiting Professor Maria Naya-Plasencia's Lab](#))
- May 2024 [IRISA](#), Rennes, France ([Visiting IRISA Lab](#))
- May 2024 [LORIA](#), Nancy, France ([Visiting LORIA Lab](#))
- Mar. 2024 [University of Hyogo](#), Kobe, Japan ([Visiting Professor Takanori Isobe's Lab](#))

Experience

- 2024–Present **Postdoctoral Researcher**, *Ruhr University Bochum*, Bochum, Germany.
Conducting research on symmetric-key cryptanalysis under the supervision of Prof. Gregor Leander in the [Symmetric Cryptography Group](#).
- 2022–2024 **Ph.D. Candidate**, *Graz University of Technology*, Graz, Austria.
Conducted research on symmetric-key cryptanalysis under the supervision of Dr. Maria Eichlseder. [Institute profile](#).
- 2021–2022 **Ph.D. Student**, *Graz University of Technology*, Graz, Austria.
Conducted research on symmetric-key cryptanalysis under the supervision of Dr. Maria Eichlseder in the [Cryptography Research Area](#).

Teaching Experience

- Summer 2023, 2024 **Guest Lecturer**, *Graz University of Technology*, Graz, Austria.
Course: Cryptanalysis.
- Fall 2016 **Teaching Assistant**, *University of Tehran*, Tehran, Iran.
Course: A Course in Cryptography.

Fall 2014 **Teaching Assistant**, *K. N. Toosi University of Technology*, Tehran, Iran.
Course: Introduction to Cryptography.

Honors

- 2019 🏆 **Winner**, *NSUCRYPTO 2019*, October 13–21.
[Competition results](#)
- 2014 🏆 **Bronze Medal**, *38th National Mathematical Competition for University Students*, Kerman, Iran, May.
[Competition website](#)
- 2013 🏆 **Bronze Medal**, *37th National Mathematical Competition for University Students*, Semnan, Iran, May.
[Competition website](#)

Computer Skills

Programming Language

Advanced PYTHON, C, C++, VHDL

Intermediate JAVA, ASSEMBLY(AVR)

Software, Tools & Packages

Math SageMath, Matlab, Maple, CoCoA

SAT PySAT, Cadical, Minisat, CryptoMinisat

SMT PySMT, Z3, STP

MILP Gurobi

CP Minizinc

Productivity L^AT_EX, Microsoft Office Tools, Texstudio

Platforms Linux, Mac, Windows

IDEs Visual Studio Code, Microsoft Visual Studio, Eclipse

Hardware/EDA Xilinx ISE, Altium Designer, PSpice, CodeVision, Atmelstudio, Arduino, Proteus

Research Interests

- Cryptanalysis and Design of Symmetric-key Cryptographic Primitives
- Automating Cryptanalysis
- Side-channel and Fault Analyses
- Logic and Boolean Algebra
- Algebra and Graph Theory
- Probability Theory and Statistics
- High-Performance and Secure Implementations

Attended Seminars/Conferences/Workshops

Mar. 2026 **FSE 2026**, Singapore, Singapore, March 23–27 (Attendee)

- Mar. 2026 **ASK 2026**, Nanyang Technological University, Singapore, March 19–22 (Invited attendee)
- Mar. 2025 **FSE 2025**, Rome, Italy, March 17–21 (**Attendee with accepted paper**)
- Mar. 2025 **SKCAM 2025**, Rome, Italy (Invited speaker)
- Dec. 2024 **ASK 2024**, TCG CREST, Kolkata, India, December 14–17 (**Invited speaker**)
- Aug. 2024 **CRYPTO 2024**, Santa Barbara, USA, August 18–22 (**Paper presentation**)
- Apr. 2024 **Beating Real-Time Crypto: Solutions and Analysis**, Lorentz Center, Leiden, Netherlands, April 22–26 (Invited attendee and speaker)
- Mar. 2024 **FSE 2024**, Leuven, Belgium, March 25–29 (**Paper presentation**)
- Apr. 2023 **EUROCRYPT 2023**, Lyon, France, April 23–27 (**Paper presentation**)
- Mar. 2023 **FSE 2023**, Kobe, Japan, March 20–24 (**Paper presentation**)
- Sept. 2022 **FRISIACRYPT 2022**, Terschelling, Netherlands, September 25–28 (Invited attendee)
- Sept. 2022 **CHES 2022**, Leuven, Belgium, September 18–21 (**Paper presentation**)
- Jun. 2022 **ACNS 2022**, Rome, Italy, June 20–23 (**Paper presentation**)
- Mar. 2022 **FSE 2022**, Athens, Greece, March 20–25 (**Paper presentation**)
- Feb. 2020 **Symmetric Cryptography in Theory and Practice**, ISC Winter School on Information Security and Cryptology (ISCwISC 2020), Iran University of Science and Technology, February 2–3 (Speaker)
- Aug. 2019 16th International ISC Conference on Information Security and Cryptology (ISCISC 2019), Ferdowsi University of Mashhad, Mashhad, Iran, August 28–29
- Jun. 2019 9th IPM-HPC Workshop on Multi-core Systems and Graphic Processors, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran, June 8–9
- Feb. 2019 8th IPM-HPC Workshop on Multi-core Systems and Parallel Platforms, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran, February 20–21
- Feb. 2019 **Selected Areas in Post-Quantum Cryptography**, Sharif University of Technology, Tehran, Iran, February 13–14
- Fall 2014 **Advanced MATLAB Workshop**, IEEE Student Branch, Faculty of Electrical Engineering, K. N. Toosi University of Technology, Tehran, Iran

Languages

Persian	Mother tongue
English	Advanced
German	Learning
Chinese	Learning

Other Information

Memberships

2021-Present International Association for Cryptologic Research
2012-2013 Iranian Mathematical Society

Interests

- Tennis
- Hiking and Mountain Climbing
- Traveling
- Reading Books
- Music (Guitar, Piano)
- Swimming
- Biking
- Movies and Documentaries
- Programming
- Puzzle-Solving and Chess

Personal Details

Languages Persian, English
Email id hsn.hadipour@gmail.com
GitHub  <https://github.com/hadipourh>
Google Scholar <https://scholar.google.com/citations?user=3gNyYaAAAAAJ&hl=en>
DBLP <https://dblp.org/pid/244/8979.html>
IACR Profile <https://www.iacr.org/cryptodb/data/author.php?authorkey=11275>

Declaration

I hereby declare that the above mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above mentioned particular.

Date: July 11, 2026

Place: Bochum, Germany

Hossein Hadipour